

When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model Reinforcement Learning

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Abstract—Reinforcement learning from rule-based verifiers (RLVR), notably Group Relative Policy Optimization (GRPO), has emerged as a cornerstone for post-training Large Language Models (LLMs) on complex reasoning tasks. A natural hypothesis is that regularizing policy gradient advantages by trajectory-level uncertainty could prevent policy collapse and filter noisy exploration traces. In this work, we conduct a systematic empirical and architectural audit of uncertainty-regularized GRPO. First, we show that Monte Carlo (MC) dropout probing becomes degenerate when the target architecture lacks active dropout in its evaluated forward graph, as confirmed on Qwen2.5. Second, through diagnostic evaluations on untouched GSM8K and SVAMP datasets (N=100), we find that internal confidence proxies—token predictive entropy ($r = +0.486$), mean token negative log-likelihood ($r = +0.432$), and logit margins ($r = +0.495$)—are strongly correlated with derivation length and arithmetic step complexity. In stress tests, internal token entropy misidentifies correct multi-step reasoning traces as more uncertain than short incorrect errors in 42.1% of paired comparisons. While external consensus disagreement (Self-Consistency) provides unconfounded offline error discrimination (AUROC = 0.812), a preregistered 5-way controlled RL experiment demonstrates that using consensus weights for online advantage scaling yields no performance improvement over standard outcome-supervised GRPO (80.00% vs 80.00%; Delta = 0.00%).

Our findings demonstrate that high offline error-predictive validity does not guarantee online reinforcement learning credit utility, establishing the necessity of negative controls when evaluating confidence-guided post-training algorithms.

Impact Statement—Reinforcement learning algorithms for Large Language Models increasingly govern automated reasoning systems in mathematics, coding, and decision support. Integrating confidence or uncertainty weights into policy gradient updates is frequently proposed to prevent reward hacking and model drift. This work provides a rigorous methodological warning: internal uncertainty signals frequently track legitimate reasoning complexity rather than error, causing naive uncertainty-weighted algorithms to suppress valid multi-step exploration. By establishing that offline error predictors do not automatically improve online policy learning, our study provides a reproducible diagnostic framework and negative-control protocol to prevent the adoption of unverified or counterproductive credit assignment mechanisms.

Index Terms—Reinforcement learning, Group Relative Policy Optimization (GRPO), Large Language Models, uncertainty estimation, self-consistency, reasoning complexity, post-training.

I. INTRODUCTION

Post-training autoregressive Large Language Models (LLMs) using reinforcement learning from verifiable rewards (RLVR) has achieved substantial milestones in mathematical derivation and formal deduction [1, 2]. Group Relative Policy Optimization (GRPO) [1] estimates baseline-free advantages by normalizing outcome rewards across a group of sampled rollouts:

$$A_{\hat{i}} = (R(y_i) - \mu_R) / (\sigma_R + \epsilon) \quad (1)$$

Because policy gradient methods can suffer from policy collapse and reward sparsity, numerous approaches hypothesize that weighting advantages by trajectory-level uncertainty $U(y_i)$ can protect the policy from destabilizing updates [3, 4].

In this paper, we report a rigorous empirical and architectural investigation into uncertainty-weighted credit assignment for GRPO. We evaluate three central questions:

1. **Architectural Validity (C1)**: Does MC-dropout probing produce meaningful stochastic representations on zero-dropout architectures?
2. **Diagnostic Validity (C2)**: Do internal token-level confidence proxies distinguish mathematical errors from legitimate reasoning complexity?
3. **Algorithmic Validity (C3)**: Does a validated offline error predictor (Self-Consistency [5]) improve online GRPO policy optimization over outcome-supervised baselines and negative controls?

II. ESTIMATOR VALIDITY ON ZERO-DROPOUT ARCHITECTURES

We audited the internal layer structure of open-weight causal language models, specifically inspecting Qwen/Qwen2.5-0.5B-Instruct (Qwen2ForCausalLM). We find that the model contains exactly **0 active nn.Dropout modules** in its attention and MLP blocks (attention_dropout = 0.0). Consequently, executing Monte Carlo dropout forward passes [6] with mc_dropout=True produces mathematically deterministic passes ($\text{Var}(\log P) = 0.0000000000$, Delta Logit = 0.0).

When normalized in advantage scaling equations with stability constant $\epsilon = 10^{-8}$, floating-point order-of-operations noise (approx 10^{-12}) results in a multiplier of $\exp(-\gamma * 10^{-4})$ approx 0.999965, producing policy updates collinear to standard GRPO ($\cos(\Delta \theta) = 1.000000$). We conclude that MC-dropout uncertainty estimation becomes degenerate when the target architecture contains no active dropout in the evaluated forward computation graph.

III. CONFIDENCE PROXIES AND REASONING COMPLEXITY

Evaluating candidate uncertainty proxies across untouched GSM8K confirmatory data ($N = 100$ independent prompt clusters, 98 degrees of freedom) reveals a pattern consistent with a reasoning-complexity confound:

- **Token Predictive Entropy** correlates strongly with sequence length ($r = +0.486$, 95% CI [+0.318, +0.627]) and equation count ($r = +0.421$).
- After controlling for completion length via partial correlation, the association between token entropy and correctness collapses from $r = -0.214$ to partial $r = -0.092$ ($p = 0.365$).
- In the **Correct-but-Complex Stress Test**, token entropy misidentifies correct multi-step reasoning traces as more uncertain than short incorrect errors in **42.1% of paired comparisons**.
- **Self-Consistency** ($U_{SC} = 1 - \text{modal count} / K$) [5] remains robust to length bias ($r = +0.114$, partial $r = -0.569$, $p = 8.1 \times 10^{-10}$, AUROC = 0.812).

TABLE I: Uncertainty Proxy Benchmark (GSM8K N=100)

Candidate Proxy	AUROC	r(Error)	r(Length)	Partial r
Self-Consistency (K=4)	0.812	+0.582	+0.114	-0.569 (p<1e-9)
Token Entropy	0.618	+0.214	+0.486	-0.092 (p=0.365)
Mean Token NLL	0.605	+0.198	+0.432	-0.081 (p=0.422)

Logit Margin	0.624	+0.226	+0.495	-0.104 (p=0.303)
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IV. PREREGISTERED CONSISTENCY-AWARE RL EXPERIMENT

To test whether validated offline self-consistency translates to an effective online policy gradient weight, we preregistered Consistency-Aware GRPO (CA-GRPO):

$$A_{\tilde{t}lde_i} = A_{\hat{t}lde_i} * (1.0 + \lambda * (c_i - c_{mean})) \tag{2}$$

where $c_i = 1$ if $extract(y_i) = \text{modal answer}$ else 0.

We benchmarked CA-GRPO against four matched controls across three independent training seeds on held-out test data evaluated under the canonical 256-token generation budget:

TABLE II: Preregistered 5-Way RL Control Matrix (N=3 Seeds)

Method	G	Held-Out Pass@1	Reward
Standard-GRPO	4	80.00 +/- 0.00%	0.12
Compute-Matched-GRPO	8	78.33 +/- 2.89%	0.26
Random-Weight-Control	4	75.00 +/- 5.00%	0.21
Permuted-Control	4	80.00 +/- 0.00%	0.29
CA-GRPO (Proposed)	4	80.00 +/- 0.00%	0.12

CA-GRPO did not improve performance over Standard GRPO in our tested configuration (Delta = 0.00%, Cohen's $d = 0.000$). Negative controls (permuted and random weighting) demonstrated that trajectory-specific advantage scaling provided no causal learning advantage over standard outcome-supervised GRPO.

V. LIMITATIONS AND CONCLUSION

Limitations: (1) RL evaluations used $N=3$ training seeds; (2) evaluations focused on mathematical reasoning (GSM8K, SVAMP) using Qwen2.5-0.5B-Instruct; (3) self-consistency requires extra inference compute ($K=4$ rollouts); and (4) findings evaluate specific linear advantage weighting.

Conclusion: Internal confidence proxies are consistent with a reasoning-complexity confound, and high offline error discrimination does not imply online RL credit utility. Rigorous negative controls and architectural audits are indispensable for sound RLVR post-training.

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